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## **LEMBAR PRAKTIKUM**

### **Simulasi Brute Force, Perbandingan Password, dan Implementasi MFA**

**Mata Kuliah:** Keamanan Jaringan

#### **A. Tujuan Praktikum**

Mahasiswa mampu:

1. Melakukan simulasi brute force di lingkungan lab
2. Membandingkan password lemah vs kuat
3. Mengimplementasikan MFA pada sistem Linux
4. Menganalisis efektivitas keamanan autentikasi

#### **B. Aturan Wajib**

- Praktikum hanya di **VM (tidak ke jaringan publik)**
- Gunakan akun dummy
- Tidak melakukan serangan ke sistem nyata

#### **C. Tools & Lingkungan**

- VM Linux (Ubuntu / Kali Linux)
- Tools:
  - hydra
  - john
  - google-authenticator
  - openssh-server

#### **D. LANGKAH PRAKTIKUM**

##### **BAGIAN 1 – SETUP**

## 1. Install tools

sudo apt update

```
root@kevin:/usr/share/wordlists# sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://archive.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
root@kevin:/usr/share/wordlists#
```

sudo apt install hydra john openssh-server libpam-google-authenticator -y

```
root@kevin:/usr/share/wordlists# sudo apt install hydra john openssh-server libpam-google-authenticator -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
john is already the newest version (1.8.0-2build1).
hydra is already the newest version (9.0-1build1).
libpam-google-authenticator is already the newest version (20170702-2).
openssh-server is already the newest version (1:8.2p1-4ubuntu0.13).
The following packages were automatically installed and are no longer required:
  libfwupdplugin1 libxmlb1
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@kevin:/usr/share/wordlists#
```

Aktifkan SSH:

sudo systemctl start ssh

```
root@kevin:/usr/share/wordlists# sudo systemctl start ssh
root@kevin:/usr/share/wordlists#
```

## BAGIAN 2 – SIMULASI BRUTE FORCE

### 2. Buat user uji

sudo adduser userlemah

Set password:

123456

```
root@kevin:/usr/share/wordlists# sudo adduser userlemah1
Adding user `userlemah1' ...
Adding new group `userlemah1' (1003) ...
Adding new user `userlemah1' (1003) with group `userlemah1' ...
Creating home directory `/home/userlemah1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for userlemah1
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
root@kevin:/usr/share/wordlists#
```

**Tambahan :**

**buat Direkoti wordlists**

`sudo mkdir -p /usr/share/wordlists`

**Masuk direktori wordlists**

`cd /usr/share/wordlists`

```
root@kevin:/usr/share/wordlists# sudo mkdir -p /usr/share/wordlists
root@kevin:/usr/share/wordlists# cd /usr/share/wordlists
root@kevin:/usr/share/wordlists#
```

**install worlists**

`sudo wget https://github.com/brannondorsey/naive-hashcat/releases/download/data/rockyou.txt`

```
-11-09&sr=b&spr=https&se=2026-05-12T12%3A58%3A41Z&rscd=attachment%3B+filename%3D
rockyou.txt&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947
aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-05-12T11%3A58%3A13Z&s
ke=2026-05-12T12%3A58%3A41Z&sks=b&skv=2018-11-09&sig=Mf8o4hFalNnHwILBndzMX%2Fj%2
FvbZVB2QYl7wu%2F4deBFw%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJna
XRodWIuY29tIiwiaXVkiIjoicmVsZWZzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5I
joia2V5MSIsImV4cCI6MTc3ODU5MTMzNywibmJmIjoxNzc4NTg3NzM3LCJwYXRoIjoicmVsZWZzZWZzc
2V0cHJvZHVjdGlvbi5ibG9iLmNvcmlud2luZG93cy5uZXQifQ.5YUni3npMlcJ_1yR1VwrMhaxAFRg-h
lgU_gyUSIANmM&response-content-disposition=attachment%3B%20filename%3Drockyou.tx
t&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent
.com)... 185.199.108.133, 185.199.110.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercon
tent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 139921497 (133M) [application/octet-stream]
Saving to: 'rockyou.txt.1'

rockyou.txt.1      100%[=====>] 133.44M  14.1MB/s   in 9.5s

2026-05-12 12:09:12 (14.0 MB/s) - 'rockyou.txt.1' saved [139921497/139921497]

root@kevin:/usr/share/wordlists#
```

### Cek file rockyou.txt di worlists

ls -lh /usr/share/wordlists/rockyou.txt

```
root@kevin:/usr/share/wordlists# ls -lh /usr/share/wordlists/rockyou.txt
-rw-r--r-- 1 root root 134M Dec  8  2021 /usr/share/wordlists/rockyou.txt
root@kevin:/usr/share/wordlists#
```

### 3. Jalankan brute force

hydra -l userlemah -P /usr/share/wordlists/rockyou.txt localhost ssh

### HASIL YANG DIHARAPKAN

- Password ditemukan cepat
- Waktu relatif singkat

```

root@kevin:/usr/share/wordlists# hydra -l userlemah -P /usr/share/wordlists/rockyou.txt localhost ssh
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-12 11:54:25
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (1:1/passwords), ~896525 tries per task
[DATA] attacking ssh://localhost:22/
[22][ssh] host: localhost login: userlemah password: 12345
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 targets did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-12 11:54:40
root@kevin:/usr/share/wordlists# █

```

### BAGIAN 3 – PASSWORD KUAT

#### 4. Ubah password menjadi kuat

sudo passwd userlemah

Gunakan:

A!b9#Xz2026

```

root@kevin:/usr/share/wordlists# sudo passwd userlemah
New password:
Retype new password:
passwd: password updated successfully
root@kevin:/usr/share/wordlists# █

```

#### 5. Jalankan brute force lagi

hydra -l userlemah -P /usr/share/wordlists/rockyou.txt localhost ssh

```

root@kevin:/usr/share/wordlists# hydra -l userlemah -P /usr/share/wordlists/rock
you.txt localhost ssh
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret
service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-12 11:56:
52
[WARNING] Many SSH configurations limit the number of parallel tasks, it is reco
mmended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip wa
iting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (l:1/p:
14344398), ~896525 tries per task
[DATA] attacking ssh://localhost:22/
[STATUS] 166.00 tries/min, 166 tries in 00:01h, 14344238 to do in 1440:12h, 16 a
ctive

```

waktu lama : 1440 jam lebih 12 menit.

## HASIL YANG DIHARAPKAN

- Password tidak ditemukan / sangat lama
- Perbedaan signifikan dibanding password lemah

## BAGIAN 4 – ANALISIS PASSWORD

**Tambahan :**

### Install Cracklib

apt install cracklib-runtime -y

```

Preconfiguring packages ...
Selecting previously unselected package libcrack2:amd64.
(Reading database ... 123151 files and directories currently installed.)
Preparing to unpack .../libcrack2_2.9.6-3.2_amd64.deb ...
Unpacking libcrack2:amd64 (2.9.6-3.2) ...
Selecting previously unselected package cracklib-runtime.
Preparing to unpack .../cracklib-runtime_2.9.6-3.2_amd64.deb ...
Unpacking cracklib-runtime (2.9.6-3.2) ...
Selecting previously unselected package wamerican.
Preparing to unpack .../wamerican_2018.04.16-1_all.deb ...
Unpacking wamerican (2018.04.16-1) ...
Setting up wamerican (2018.04.16-1) ...
Setting up libcrack2:amd64 (2.9.6-3.2) ...
Setting up cracklib-runtime (2.9.6-3.2) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.18) ...
root@kevin:/usr/share/wordlists#

```

## 6. Uji kekuatan password

cracklib-check

Uji:

123456

```
root@kevin:/usr/share/wordlists# echo "12345" | cracklib-check
12345: it is too short
root@kevin:/usr/share/wordlists#
```

admin123

```
root@kevin:/usr/share/wordlists# echo "admin123" | cracklib-check
admin123: it is based on a dictionary word
root@kevin:/usr/share/wordlists#
```

A!b9#Xz2026

```
root@kevin:/usr/share/wordlists# echo 'A!b9#Xz2026' | cracklib-check
A!b9#Xz2026: OK
root@kevin:/usr/share/wordlists#
```

hasil ok : karena passwornya ini

### CATAT:

- Password lemah → mudah ditebak
- Password kuat → ditolak / dianggap kuat

## BAGIAN 5 – IMPLEMENTASI MFA

**Tambahan :**

**Masuk ke user lemah**

su - userlemah

```
root@kevin:/usr/share/wordlists# su - userlemah
userlemah@kevin:~$ google-authenticator
```

## 7. Setup MFA

google-authenticator

Ikuti instruksi:

- Simpan secret key
- Simpan emergency codes

```
Your new secret key is: TGWGRMZQT4F53GTIJEIW54E5JE
Your verification code is 703263
Your emergency scratch codes are:
  35544809
  10951319
  56362107
  11256219
  58799376
```

```
Do you want me to update your "/home/userlemah/.google_authenticator" file? (y/n) y

Do you want to disallow multiple uses of the same authentication token? This restricts you to one login about every 30s, but it increases your chances to notice or even prevent man-in-the-middle attacks (y/n) y

By default, a new token is generated every 30 seconds by the mobile app. In order to compensate for possible time-skew between the client and the server, we allow an extra token before and after the current time. This allows for a time skew of up to 30 seconds between authentication server and client. If you experience problems with poor time synchronization, you can increase the window from its default size of 3 permitted codes (one previous code, the current code, the next code) to 17 permitted codes (the 8 previous codes, the current code, and the 8 next codes). This will permit for a time skew of up to 4 minutes between client and server.
Do you want to do so? (y/n) y

If the computer that you are logging into isn't hardened against brute-force login attempts, you can enable rate-limiting for the authentication module. By default, this limits attackers to no more than 3 login attempts every 30s.
Do you want to enable rate-limiting? (y/n) y
userlemah@kevin:~$
```

## 8. Aktifkan MFA di SSH

Edit:

```
sudo nano /etc/pam.d/sshd
```

Tambahkan:

```
auth required pam_google_authenticator.so
```



```
GNU nano 4.8 /etc/pam.d/sshd
# PAM configuration for the Secure Shell service

# Standard Un*x authentication.
@include common-auth

auth required pam_google_authenticator.so

# Disallow non-root logins when /etc/nologin exists.
account      required      pam_nologin.so
```

Edit config SSH:

```
sudo nano /etc/ssh/sshd_config
```

Ubah:

ChallengeResponseAuthentication yes

```
# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
ChallengeResponseAuthentication yes
```

Restart SSH:

```
sudo systemctl restart ssh
```

```
root@kevin:/usr/share/wordlists# sudo systemctl restart ssh
root@kevin:/usr/share/wordlists#
```

**Tambahan :**

**Hubungkan Authenticator di hp atau di ekstensi untuk tahu otp nya**

Edit Account

×

Account Name

ubuntu

Service

ubuntu

Secret Key

TGWGRMZQT4F53GTIJEIW54E5JE

🗕

Cancel

Save

## 9. Uji Login MFA

ssh userlemah@localhost

```
root@kevin:/home/kevin# ssh userlemah@localhost
Password:
Verification code:
Password: █
```



Authenticator

   

🔍

Search accounts...

ubuntu: ubuntu

826 202

5

```
For more details see:
https://ubuntu.com/20-04

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

178 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue May 12 12:50:21 2026 from 127.0.0.1
userlemah@kevin:~$
```

Akan diminta:

- Password
- OTP (kode dari authenticator)

## E. ANALISIS WAJIB

Jawab:

1. Berapa lama brute force menemukan password lemah?

jawab : Kurang dari 5 menit karena bergantung urutan list password di file rockyou.txt. semakin urutan berada diawal maka akan semakin mudah ditemukan (12345 berurutan jadi berada pada urutan paling awal berdasar kombinasi angka dan huruf)

2. Mengapa password kuat sulit ditemukan?

karena pakai kombinasi angka, huruf, dan karakter sehingga sulit atau bahkan tidak ditemukan di rockyou.txt sehingga sulit dideteksi menggunakan brute force dan bahkan membutuhkan waktu lama atau tidak berhasil dideteksi.

3. Apa perbedaan hasil antara dua pengujian?

jika menggunakan password 12345 sangat mudah ditemukan bahkan ketika uji coba cracklib hasilnya it is too short sehingga mudah ditemukan. berbeda ketika menggunakan password A!b9#Xz2026 maka akan membutuhkan waktu lama untuk dideteksi brute force yaitu 12 jam lebih karena password menggunakan kombinasi yang sulit.

4. Apakah brute force masih berhasil setelah MFA?

tidak, walaupun brute force berhasil mengetahui kata sandinya tetapi masih ada MFA yang menggunakan kode yang hanya bisa diakses oleh si user yang memiliki secret key authenticator

5. Mengapa MFA meningkatkan keamanan?

karena MFA menggunakan secret key dan perlu otp untuk mengaksesnya. sehingga jika tidak punya secret key yang diakses melalui authenticator maka akan sulit di deteksi asswordnya

**Kesimpulan :** Brute Force merupakan cara untuk mendeteksi password. semakin lemah password (12345) maka akan semakin mudah untuk ditemukan dan waktu menemukannya pun juga singkat. jika password menggunakan kombinasi angka, huruf, dan karakter maka akan semakin sulit juga untuk dideteksi dan membutuhkan waktu lama. jika ingin lebih aman bisa menggunakan mfa berbasis otp. walaupun brute force berhasil, tetapi jika ada mfa maka akan lebih aman.

## **F. FORMAT LAPORAN**

1. Tujuan
2. Langkah praktikum
3. Screenshot hasil:
  - brute force (lemah vs kuat)
  - password test
  - MFA login
4. Analisis
5. Kesimpulan